Linear Dependence, Independence, Basis, and Dimension

Lecture Notes

1 Linear Combination

Definition

Let V be a vector space over a field $\mathbb{F}$. Vectors $v_1, v_2, \ldots, v_n \in V$.

An expression

$$a_1v_1 + a_2v_2 + \cdots + a_nv_n, \quad a_i \in \mathbb{F},$$

is called a **linear combination** of the vectors.

Example 1

Let

$$v_1 = (1, 2), \quad v_2 = (3, -1).$$

Then

$$2v_1 - v_2 = 2(1, 2) - (3, -1) = (2, 4) - (3, -1) = (-1, 5).$$

2 Linear Dependence and Independence

Definition (Linear Dependence)

Vectors $v_1, v_2, \ldots, v_n$ are **linearly dependent** if there exist scalars, not all zero, such that

$$a_1v_1 + a_2v_2 + \cdots + a_nv_n = 0.$$

Definition (Linear Independence)

Vectors $v_1, v_2, \ldots, v_n$ are **linearly independent** if

$$a_1v_1 + a_2v_2 + \cdots + a_nv_n = 0 \Rightarrow a_1 = a_2 = \cdots = a_n = 0.$$

3 Numerical Examples

Example 2: Two Vectors in $\mathbb{R}^2$

Check whether

$$v_1 = (1, 2), \quad v_2 = (2, 4)$$

are linearly independent.

Solution:

$$a(1,2) + b(2,4) = (0,0)$$

$$\Rightarrow \begin{cases} a + 2b = 0 \\ 2a + 4b = 0 \end{cases}$$

Non-trivial solutions exist. Hence, the vectors are **linearly dependent**.

Example 3: Standard Basis Vectors

Check whether $(1,0)$ and $(0,1)$ are linearly independent.

Solution:

$$a(1,0) + b(0,1) = (0,0) \Rightarrow a = 0,\ b = 0.$$

Hence, the vectors are **linearly independent**.

Example 4: Three Vectors in $\mathbb{R}^3$

Check linear independence of

$$(1,0,2),\ (2,1,3),\ (3,1,5).$$

Solution:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 2 & 3 & 5 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0$$

Row reduction yields a free variable, so a non-trivial solution exists.

$$\Rightarrow \textbf{Linearly dependent}$$

Example 5: Polynomial Space

Check whether $\{1 + x,\ 1 - x,\ 2\}$ is linearly independent in P_1.

Solution:

$$a(1 + x) + b(1 - x) + c(2) = 0$$

Comparing coefficients:

$$\begin{cases} a + b + 2c = 0 \\ a - b = 0 \end{cases}$$

Hence, the set is **linearly dependent**.

4 Important Properties

- Any set containing the zero vector is linearly dependent.
- In $\mathbb{R}^n$, any set of more than n vectors is linearly dependent.
- Any subset of a linearly independent set is linearly independent.

5 Basis of a Vector Space

Definition

A **basis** of a vector space V is a set of vectors that

(i) is linearly independent,

(ii) spans V.

Example 6: Standard Basis of $\mathbb{R}^3$

$$\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

This is a basis of $\mathbb{R}^3$.

Example 7: Non-Standard Basis

Check whether

$$\{(1, 1), (1, -1)\}$$

is a basis for $\mathbb{R}^2$.

Solution:

$$\det \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = -2 \neq 0.$$

Hence, it is a basis.

Example 8: Basis of Polynomial Space

A basis of P_2 is

$$\{1, x, x^2\}.$$

6 Dimension of a Vector Space

Definition

The **dimension** of a vector space is the number of vectors in any basis of the space.

Example 9

$$\dim(\mathbb{R}^n) = n.$$

Example 10

$$\dim(P_3) = 4, \quad \text{basis } \{1, x, x^2, x^3\}.$$

Example 11: Matrix Space

The dimension of $M_{2\times 2}$ is 4.

7 Practice Problems

1. Check whether $(1, 2, 3), (2, 4, 6), (3, 6, 9)$ are linearly independent.
2. Find the dimension of the space spanned by
$$\{(1, 0, 1), (2, 1, 3), (3, 1, 4)\}.$$
3. Find a basis for the solution space of
$$x + y + z = 0.$$
4. Find the dimension of the space of symmetric 3×3 matrices.

Answers (Brief)

- Linearly dependent
- Dimension $= 2$
- Basis: $\{(-1, 1, 0), (-1, 0, 1)\}$
- Dimension $= 6$